15th of January 2026 Full Name: _____

Question	Points	Score
1	4	
2	4	
3	4	
Total:	12	

PRACTICE EXAM

Multiple Choice Questions

1. Each question in this section may have one or more correct answers. Select your answers by marking the corresponding boxes in the **Answer Grid** below. **Only marks inside of the Answer Grid are considered.** (4)

   - You receive **one point** for each correctly selected option.
   - You lose **half a point** for each incorrectly selected option.
   - If a correct option is not selected, **no points are awarded or deducted**.

   Corrections: You may change an answer by circling the desired box(es) in the Answer Grid. If at least one box is circled for a question, then:

   - all non-circled marks for that question are ignored, and
   - only the circled box(es) are considered during grading.

   Answer Grid
   For each question part (a, b, c, ...), mark the corresponding option(s) A–D.

Q	A	B	C	D
(a)	☐	☐	☐	☐
(b)	☐	☐	☐	☐
(c)	☐	☐	☐	☐

PRACTICE EXAM

(a) What is the default naming convention for a RSA-based SSH private key?

A. `rsa.pvt`
B. `id_rsa.pvt`
C. `id_rsa.pub`
D. `id_rsa`

(b) Check all correct statements, assuming no protection mechanisms (such as firewalls) are in place. Nmap considers a `X` port that does not answer connection requests `Y`.

A. `X` = TCP, `Y` = open|filtered
B. `X` = TCP, `Y` = closed
C. `X` = UDP, `Y` = closed
D. `X` = UDP, `Y` = open|filtered

(c) Which of the following properties is/are taken into account to decide whether an NSE (Nmap Scripting Engine) script should be part of the default category? Select all answers that apply.

A. Speed
B. Customizable Settings
C. Modularity
D. Intrusiveness

PRACTICE EXAM

Open Questions

2. In this section of the exam, you will encounter open-ended questions. Please read each question carefully and answer as precisely as possible. To achieve the best results, follow these guidelines:

   - **Accuracy**: Ensure you fully understand each question and formulate your answer to directly respond to it.
   - **Write on the provided lines**: Please write your answers on the provided lines.
   - **Clarity and legibility**: Write neatly and ensure your handwriting is readable. Illegible answers cannot be graded. Unclear handwriting may result in point deductions.
   - **Grammatical correctness**: While complete and grammatically correct sentences are preferred, they are not mandatory. The key is that your answers are clear and understandable. As long as your intent is clear, your answer will be considered valid.
   - **Comprehensive and concise answers**: Answer as concisely as possible and avoid including irrelevant information.

PRACTICE EXAM

(a) Explain the difference between reverse and bind shells. Furthermore, elaborate on which of these two approaches is generally more stealthy when dealing with a standard firewall. (2)

.....
.....
.....
.....

(b) Explain the main difference between encryption, encoding, and hashing.

.....
.....
.....
.....
.....
.....

(c) Explain the difference between vertical and horizontal privilege escalation. (2)

.....
.....
.....
.....

PRACTICE EXAM

Technical Analysis and Interpretation

This section assesses your ability to analyze technical artifacts, reason about their behavior, and derive correct responses or actions based on given constraints.

If a task consists of multiple subtasks, and unless explicitly stated otherwise, the subtasks build upon each other and information obtained in earlier subtasks may be used in subsequent ones.

PRACTICE EXAM

Web Exploitation - SQL Injection

3. You are analyzing a web application running on a Linux system with Apache and PHP, which queries a MySQL database. The application's root directory is `/var/www/html`.

   The authentication mechanism relies on a PHP script querying the database for user credentials, similar to examples covered in the lecture.

   In the following tasks, you have to craft SQL injection payloads to authenticate yourself without knowing any valid credentials.

   - Provide your payload character by character in the given boxes.
   - Given a final answer, cross out any trailing boxes that are unused. An empty box represents a whitespace.
   - Use one row for your answer. If you wish to revise your answer, use the next row and strike through the previous attempt to mark it as invalid.

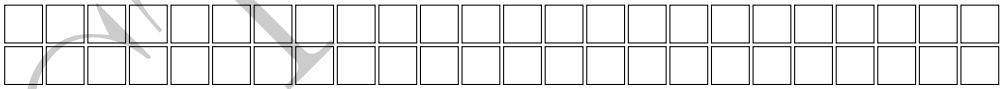
   (a) The PHP code in Codeblock 1 defines the SQL query used in the backend: (2)

```
1 $sql_query = "SELECT * FROM users WHERE username = '$user'
    AND password = '$pass'";
```

Codeblock 1: SQL Query 1

Conditions:

- You must inject using the `user` variable.
- You must end the payload with a line comment. Note, that the line comment syntax in MySQL is `--` .
- You know, that a user with the username `chomsky` exists.



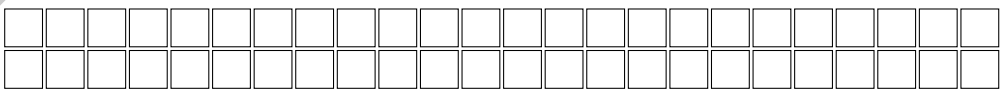
   (b) The PHP code in Codeblock 2 defines the SQL query used in the backend: (2)

```
1 $sql_query = "SELECT * FROM users WHERE password = '$pass'
    AND username = '$user'";
```

Codeblock 2: SQL Query 2

Conditions:

- You must inject using the `user` variable.



PRACTICE EXAM